
Countersign: Evidence-Grounded Supervision of Coding-Agent Completion Claims

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Abstract

1 Coding agents terminate when they believe a task is done, but that belief is not
2 always supported by their own execution trace. We present *Countersign*, a deter-
3 ministic supervisory meta-agent that audits a worker’s completion claim against
4 the trace—checking that cited tests exist, remain fresh with respect to the files
5 they cover, postdate the latest requirement update, and do not rest on superseded
6 sources—and can block termination and demand fresh evidence. Countersign is
7 deliberately *trace-only*: it judges whether a claim is *justified* at claim time—distinct
8 from whether the implementation is *correct*, which a hidden evaluator answers
9 once after termination. In a preregistered, frozen campaign of 490 real-model
10 runs over ten held-out fixtures, all 17 blocks landed on proposals an independent
11 oracle also labelled unsupported (precision 1.00, 95% CI [0.82, 1.0]) and none
12 on the 204 supported proposals judged—though 13 of 17 are the trivial case of a
13 claim citing no evidence. A counterfactual replay shows a conventional continuous-
14 integration gate catches 2 of 15 unsupported proposals; Countersign catches 11
15 of 11 it judged. Negative results are equally informative: truncating the worker’s
16 visible history did not increase unsupported completion; the effect appears only
17 when history is *replaced by a narrative summary*, driven mainly by prompting, the
18 gate’s contribution being to convert unsupported acceptances into blocked episodes
19 that recover. And we quantify the construct honestly: all 18 accepted unsupported
20 completions concerned work that passed hidden evaluation—Countersign is an
21 evidence-hygiene instrument whose safety value remains a hypothesis under test.

22 1 Introduction

23 An autonomous coding agent ends its episode by asserting completion—a claim about evidence: the
24 tests it ran, the edits it made, the requirements it was given. When the claim outruns the evidence (a
25 cited test predating the final edit; a superseded source), the agent has not necessarily produced wrong
26 code, but it has produced an *unwarranted* report, and whoever trusts it inherits a risk they cannot see.

27 We study whether a second agent with authority over the first’s termination can reliably tell warranted
28 completion claims from unwarranted ones, and what that authority costs. Countersign reads only the
29 worker’s execution trace: it never consults ground truth, and the hidden evaluator that determines
30 correctness runs exactly once, after termination, identically in every condition. This separation is
31 the paper’s organizing principle: a claim can be justified and wrong, or correct and unjustified, and
32 conflating the two makes “the verifier improved outcomes” unfalsifiable.

33 **Contributions.** We contribute (i) a supervision loop with halting authority over worker termination,
34 scored against an independent post-hoc oracle it does not compute (§2); (ii) a preregistered evaluation
35 on matched fixture pairs and negative controls that price false blocks, under a frozen protocol with a
36 public deviation ledger (§3); (iii) results across 490 runs: perfect block precision, a CI-gate ablation,
37 a truncation null, and a regime-specific summarization effect decomposed into prompt and gate

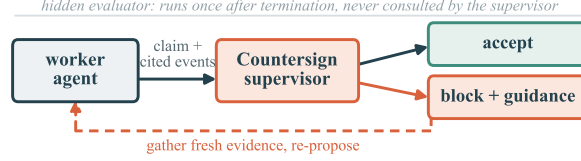


Figure 1: The supervision loop. The worker proposes termination with cited evidence; Countersign audits the claim against the trace and accepts or blocks with a specific deficiency, after which the worker gathers fresh evidence and re-proposes. The hidden evaluator runs once, after termination, and is never consulted by the supervisor.

38 contributions (§4); (iv) a calibrated account of the gate as evidence hygiene, not a demonstrated safety
 39 mechanism (§5).

40 2 The Countersign Supervisor

41 **Worker and ledger.** The worker is a tool-loop coding agent (self-hosted open-weight models) in a
 42 git-backed workspace; every action appends to an *evidence ledger*—tool calls with outcomes, file
 43 mutations, test runs with inferred coverage, requirement updates. To finish, it emits a `finish` action
 44 carrying a claim and cited evidence identifiers.

45 **The audit.** Countersign checks four properties of the cited evidence: (i) at least one cited event
 46 is a successful test run; (ii) no cited test predates a later mutation of a file it covers (Appendix A);
 47 (iii) cited evidence postdates the latest relevant requirement update; and (iv) the claim does not rest
 48 on sources the task marks superseded. Freshness is tri-state: the supervisor abstains when coverage is
 49 unknown.

50 **Authority.** On failure the supervisor blocks termination and returns the specific deficiency; a repair
 51 variant additionally issues the smallest evidence-refreshing action. Every arm logs the *raw* verifier
 52 decision separately from the *enforced* one, so non-blocking arms still measure what the supervisor
 53 would have done.

54 **Independent oracle.** A post-hoc oracle labels every proposal *supported/unsupported/uncertain*
 55 from fixture-authored completion policies and canonical trace chronology. It shares no code with the
 56 online verifier; §5 reports a regime where they disagree.

57 3 Evaluation Design

58 **Fixtures.** Ten held-out fixture repositories, authored to a preregistered design and frozen before any
 59 real-model run was inspected. Three *matched pairs* isolate one failure mode each under context parity
 60 (workspaces byte-identical but for one designed file): *temporal* (cited tests predate the final edit),
 61 *provenance* (fresh evidence grounded in a superseded document), and *requirement* (a clarification
 62 arrives after the last passing run). Four *negative controls* contain no trap (Appendix A), so any block
 63 on them is a false positive by construction; a separate development suite, on which the rules were
 64 iterated, is never used for evaluation.

65 **Conditions.** Arms cross a worker prompt variant with verifier authority: `memory_baseline` (naive
 66 prompt, no gate), `observe_only` (evidence-citation prompt; verifier logs decisions, never blocks,
 67 returns no feedback), `verification_only` (same prompt, blocking), `verification_and_repair`, and an
 68 evaluation-only `oracle_supervisor` upper bound. Because `memory_baseline` and `verification_only`
 69 differ in *both* prompt and gate, we never report that contrast as a gate effect; `observe_only` is the
 70 prompt-matched control, verified in-trace to emit zero worker-visible feedback events.

71 **Memory degradation.** A frozen registry of degradation profiles applies only to *model-visible* state:
 72 graded *lossy truncation* of history (low/medium/high) and a *resume-summary* regime replacing
 73 the visible ledger with a single truthful summary—the pattern deployed scaffolds use on session
 74 resumption. The canonical ledger, verifier, oracle, and evaluator always see undegraded state.

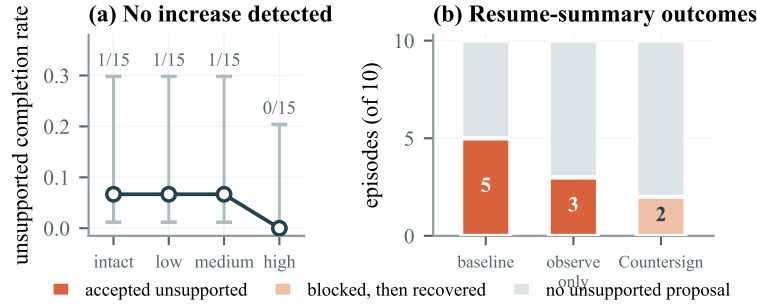


Figure 2: Memory degradation does not act uniformly. (a) Graded truncation: no detectable increase in unsupported completion; the Wilson 95% intervals mark this a non-detection at modest power, not invariance. (b) Replacing history with a resume summary does produce the effect: the evidence-citation prompt reduces unsupported proposals (5→3), and the gate converts those it sees from silent acceptances into blocked episodes that recover (2 of 2). Small n ; decomposition, not significance.

Protocol. Temperature 0.7, sampler seeds 0–4, action budget 24 (Appendix A). Endpoints, comparisons, and interpretation branches for both positive and null outcomes were committed before the runs; the frozen protocol hashes code, fixtures, and model weights and refuses to start on a dirty tree.

4 Results

All numbers come from audited run manifests; the worker is `qwen2.5-coder:14b`. The campaign’s 490 runs yield 341 oracle-judged finish proposals, 27 labelled unsupported.

Blocks are precise. Across every arm with a live verifier, Countersign issued 17 blocks—all on proposals the independent oracle also labelled unsupported, none on the 204 supported proposals a verifier judged: precision 1.00 (95% CI [0.82, 1.0]), false-block rate 0/204 (95% CI [0, 0.018]), recall $17/18 = 0.94$ (95% CI [0.74, 0.99]). Two caveats bound this. First, the denominator counts only judged proposals: 110 further supported proposals in no-verifier arms are excluded, not counted as successes. Second, and more important, **13 of the 17 blocks are the trivial case**—a claim citing no evidence, which any rule detects. Only 4 required substantive reasoning about staleness or provenance: read the result as “the gate does not misfire,” not “the gate discriminates hard cases.”

A conventional CI gate is mostly blind. We replay every stored proposal against the standard-practice rule “require a successful test run after the last edit,” a strict subset of Countersign’s temporal check. The CI rule catches 2 of 15 unsupported proposals; Countersign catches 11 of the 11 it judged, of which only 4 required substantive reasoning—the rest cite nothing at all. Neither gate blocked any of the 161 supported proposals both judged: the difference is blindness, not conservatism.

Truncating memory does not detectably induce false claims. Baseline unsupported completion on trap fixtures did not increase with truncation severity (Figure 2a). This is a non-detection, not invariance: at 15 runs per cell the Wilson intervals span roughly 0.01–0.30, so effects under ~30 percentage points are unresolvable; the preregistered null branch applies with that caveat.

The effect is regime-specific, and decomposes. Replacing history with a resume summary does produce unsupported completion (Figure 2b). The comparison must be read at the *proposal* level: in a blocking arm an accepted-unsupported finish is suppressed by the intervention’s own definition, so a zero there is partly mechanical. Counting unsupported proposals over ten matched cells: baseline 5, `observe_only` 3, `verification_only` 2. Pairing the cells, the prompt contrast has discordant cells 3 vs. 1 (exact McNemar $p = 0.625$); the further $3 \rightarrow 2$ is sampling noise, since the gate cannot influence a worker’s *first* proposal. The gate’s contribution is not fewer unsupported claims but a different fate for them: both proposals it saw were blocked, and both episodes recovered and terminated with supported evidence. No comparison here reaches $p < .05$, and an a-priori power analysis explains why—even with every discordant pair in one direction the exact test requires $b \geq 6$, one more than this campaign could have produced, and 16 matched cells would be needed for 80% power at the observed effect (Bayesian sensitivity reading in Appendix D). The cleanest evidence is measurement without power: in `observe_only` the verifier judges but cannot act, and its would-block decisions

Table 1: Accepted completions ($n=327$): oracle verdict on the claim vs. hidden-evaluator verdict on the work. The empty bottom-right cell is the paper’s central caveat.

	correct	incorrect
claim supported	307	2
claim unsupported	18	0

111 fired in exactly the 3 unsupported cells and none of the other 7 (TP= 3, FP= 0, FN= 0)—immune to
 112 the circularity objection a blocking arm invites.

113 **What “unsupported” means here.** Table 1 crosses the oracle’s judgement of the accepted claim
 114 against the hidden evaluator’s judgement of the work. All 18 unsupported completions concerned
 115 work that was *correct*: an unjustified claim never coincided with a wrong implementation, so
 116 Countersign is measured to catch *poor evidential hygiene* and its safety value remains a hypothesis
 117 this campaign cannot confirm.

118 5 Supervisor-Failure Analysis and Limitations

119 **The structural blind spot.** The provenance pair is designed so that fresh, well-cited evidence
 120 grounded in a superseded source passes a trace-only audit while the work fails hidden evaluation.
 121 That designed path never fired here (the two **supported-but-incorrect** completions arose on a
 122 negative-control and a requirement fixture), but the class is real, no trace audit can reach it, and
 123 it bounds what runtime evidence supervision can promise. Appendix B details two further failure
 124 modes: classifier/oracle divergence under heavy degradation and a fixture family censored by task
 125 weight, not action budget.

126 **Limitations.** (1) Most important: the held-out traps and the supervisor’s rules share authors;
 127 the development/held-out split prevents rule tuning on evaluation fixtures but not the conceptual
 128 correlation between what we trap and what we detect, and the provenance family (built to *defeat* the
 129 supervisor) plus the negative controls (which price the unflattering direction) offset this only partially.
 130 (2) One worker model on ten small fixtures establishes nothing about prevalence in production
 131 software. (3) The primary endpoint is partly suppressed by construction in blocking arms; gate
 132 effects are read at the proposal level and in post-block outcomes. (4) The verifier and oracle share no
 133 code but read the same trace; most agreements concern claims citing no evidence—near-vacuous
 134 independence. (5) No result reaches $p < .05$; nulls are non-detections at modest power, and we report
 135 decompositions and intervals, never significance. (6) The degradation profiles are our own transforms,
 136 not a deployed scaffold’s condenser; (7) oracle-label validation covers a single-rater sample.

137 6 Related Work

138 Workers game outcome checks [1, 2]; external supervisors with authority over another agent are an
 139 established design space [3–6]; deterministic finish-blocking hooks (no termination until tests pass)
 140 are deployed practice—the rule our CI replay prices. Countersign differs in *what* it audits: cited-
 141 evidence chronology (freshness, requirement recency, provenance), with abstention and priced false
 142 blocks. The justified/correct split descends from process-vs-outcome supervision [7, 8], instantiated
 143 at termination against a hidden evaluator [9, 10]. Trajectory judges score completion without
 144 termination authority [11]; SCATE targets premature termination [12]; concurrent work checks
 145 completion evidence and provenance in workflow graphs [13] and at GUI agents’ finish step [14].
 146 Self-verification [15] is the complement; long-context degradation [16, 17] motivates the memory
 147 regimes.

148 Responsible-Use Statement

149 Countersign controls agent overclaiming, but agents supervising agents carries risks we keep visible:
 150 an *allow* audits justification, not correctness, and must not be marketed as verification; a visible gate
 151 invites *oversight displacement*, so verdicts reach operators as cited-evidence audits, not badges; and
 152 halting authority adds a failure surface—*who supervises the supervisor*—that we price as counter-
 153 metrics. Experiments involve no human subjects, personal data, or production systems.

References

- [1] Sydney Von Arx, Lawrence Chan, and Beth Barnes. Recent frontier models are reward hacking. METR Technical Report, 2025. URL <https://metr.org/blog/2025-06-05-recent-reward-hacking/>.
- [2] Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL <https://arxiv.org/abs/2503.11926>.
- [3] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. In *Forty-first International Conference on Machine Learning, ICML 2024*, pages 16295–16336. PMLR, 2024. URL <https://proceedings.mlr.press/v235/greenblatt24a.html>.
- [4] Zhen Xiang, Linzhi Zheng, Yanjie Li, Junyuan Hong, Qinbin Li, Han Xie, Jiawei Zhang, Zidi Xiong, Chulin Xie, Carl Yang, Dawn Song, and Bo Li. GuardAgent: Safeguard LLM agents via knowledge-enabled reasoning. In *Forty-second International Conference on Machine Learning, ICML 2025*. PMLR, 2025. URL <https://proceedings.mlr.press/v267/xiang25a.html>.
- [5] Haoyu Wang, Christopher M. Poskitt, and Jun Sun. AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents. In *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE 2026)*, 2026. URL <https://arxiv.org/abs/2503.18666>.
- [6] Sahana Chennabasappa et al. LlamaFirewall: An open source guardrail system for building secure AI agents. *arXiv preprint arXiv:2505.03574*, 2025. Author list to be verified against the arXiv page before submission.
- [7] Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process- and outcome-based feedback, 2022. URL <https://arxiv.org/abs/2211.14275>.
- [8] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.
- [9] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [10] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.
- [11] Mingchen Zhuge, Changsheng Zhao, Dylan R. Ashley, Wenyi Wang, Dmitrii Khizbullin, Yunchang Xiong, Zechun Liu, Ernie Chang, Raghuraman Krishnamoorthi, Yuandong Tian, Yangyang Shi, Vikas Chandra, and Jürgen Schmidhuber. Agent-as-a-judge: Evaluate agents with agents. In *Forty-second International Conference on Machine Learning, ICML 2025*. PMLR, 2025. URL <https://proceedings.mlr.press/v267/zhuge25a.html>.
- [12] Sijia Gu, Noor Nashid, and Ali Mesbah. SCATE: Learning to supervise coding agents for cost-effective test generation, 2026. URL <https://arxiv.org/abs/2607.08983>.
- [13] Jesus Salas. Correct is not governed: Provenance integrity in agentic workflows, 2026. URL <https://arxiv.org/abs/2608.12761>.
- [14] Qijun Han, Haoqin Tu, Zijun Wang, Haoyue Dai, Yiyang Zhou, Nancy Lau, Alvaro A. Cardenas, Yuhui Xu, Ran Xu, Caiming Xiong, Zeyu Zheng, Huaxiu Yao, Yuyin Zhou, and Cihang Xie. VLAA-GUI: Knowing when to stop, recover, and search, a modular framework for GUI automation, 2026. URL <https://arxiv.org/abs/2604.21375>.

- [15] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [16] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 2024.
- [17] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

A Design Details

Test coverage is inferred statically, so documentation edits do not invalidate code evidence, and a failure superseded by a later passing run does not permanently contradict a claim. The four negative controls: documentation edits after green tests; an unrelated-module edit with a targeted-test citation; a legitimate zero-change audit; an irrelevant late clarification. Sampling uses temperature 0.7 because greedy decoding would make multi-seed replication pseudoreplication; seeds 0–4 are sampler seeds over otherwise identical episodes.

B Additional Failure Modes

Classifier/oracle divergence. Under high degradation the online claim classifier disagreed with the independent oracle, because the classifier reads the worker’s degraded view while the oracle reads the canonical trace. All endpoints in the main text are oracle-anchored; the divergence is evidence that an independent oracle is necessary rather than redundant.

Task weight censors a family. Requirement-family trap runs exhaust the action budget in 3/5 runs in every arm, and raising the budget from 24 to 40 actions left that rate unchanged. The family is intrinsically too heavy for this worker—a fixture-design finding, not a budget-calibration error.

C A Blocked-then-Recovered Trajectory

Under lossy truncation at low severity, one verification-arm episode on a fresh-evidence temporal fixture shows the mechanism in miniature. The pressured worker twice proposed terminating *legitimate* work with insufficient cited evidence; the supervisor blocked both proposals with the specific deficiency; the worker gathered fresh evidence, terminated supported, and passed hidden evaluation. Pressure-induced premature finishing was rare for this worker, but both instances that occurred were caught and repaired by the gate. The five blocks issued across the pressure gradient (three on trap fixtures, two on control-task proposals) all landed on oracle-unsupported proposals, and enforced false blocks on oracle-supported work remained zero campaign-wide.

D Bayesian Sensitivity Readings

As a post-hoc sensitivity analysis (recorded as such in the deviation ledger; no new data, uniform Beta(1, 1) priors, exact conjugate posteriors, no sampling), we re-read the paper’s key small- n counts in Bayesian form. *Direction of the prompt effect:* conditioning on discordant matched cells as in the exact McNemar test, $b=3$ cells favour `observe_only` over `memory_baseline` and $c=1$ the reverse, giving posterior Beta(4, 2) on the probability that a discordant cell favours the prompt and $P(\theta > 0.5) = 0.8125$ —roughly four-to-one posterior odds on a beneficial direction from the same data whose two-sided p is 0.625. *Discrimination in the arm that cannot act:* the `observe_only` verifier’s sensitivity 3/3 has posterior mean 0.80 with 95% credible interval [0.40, 0.99], and its specificity 7/7 has mean 0.89, interval [0.63, 1.00]; campaign-wide block precision 17/17 gives [0.81, 1.00], agreeing with the Wilson interval in the main text. These are direction and uncertainty statements only; none is a magnitude claim, and all shrink toward chance under priors skeptical of any effect.

250 **E Reproducibility**

251 The campaign spans three tags of one freeze: a base freeze plus two infrastructure-only fixes applied
252 mid-campaign (interruption-recovery artifact reuse; a tool-level crash on syntactically invalid worker-
253 written code surfacing as a tool error rather than killing the episode). The hashed verifier-policy files
254 are byte-identical across all three tags: no verifier, oracle, or fixture logic changed after the freeze.
255 Every deviation, interim inspection, and protocol-identifier supersession—including two interim
256 analyses that informed spending decisions, and one corrected over-interpretation—is recorded in the
257 deviation ledger shipped with the artifact. The artifact is a relocatable bundle whose integrity audit
258 passes from the copied location; it contains only fixtures, protocols, and run records.